

Professional Experience

Sage Bionetworks

(REMOTE) SEATTLE, UNITED STATES OF AMERICA

Chief Data Officer - Sage Bionetworks

March 24 –

- **Portfolio & Financial Management:** Direct accountability for a \$40M+ annual project portfolio spanning 50+ concurrent initiatives across federal agencies (NIH: NIA, NCI, NINDS, FNIH, NIMH, NHLBI, NCATS, ARPA-H), foundations (Gates Foundation, AACR, Children's Tumor Foundation, N-TAP, RWJF, Helmsley Trust, GFF), and commercial partnerships (pharmaceutical and industry contracts with AACR, Wellcome Leap, DiME). Manage diversified funding mix demonstrating sophisticated financial planning and resource allocation across multiple funding cycles. Lead commercial contract negotiations with industry partners, establishing sustainable revenue streams and partnership models that balance mission-driven research with financial sustainability.
- **Leadership & Team Management:** Lead Sage's largest team of 50+ professionals (Science Data), comprising director-level scientists, clinical-bioinformatics experts, software developers, and technical project managers across seven specialised sub-teams. Developed performance frameworks and career progression pathways, achieving strong retention in a competitive market. Built and scaled a multi-petabyte infrastructure (3.7PB) for storage, curation, and accessibility of biomedical data
- **Strategic Partnerships Business Development:** Established strategic partnerships with major technology providers and research institutions to enhance data infrastructure capabilities. Secured new funding through competitive grant applications and industry partnership proposals. Maintained high external visibility through conferences, committees, and publications, positioning Sage as a leader in biomedical data infrastructure.
- **Strategic Partnerships Business Development:** Spearheaded creation of dynamic resource allocation models to optimise delivery across the diverse scientific portfolio. Implemented standardised operating procedures and governance frameworks, improving project dissemination and cross-institutional collaboration. Delivered essential infrastructure supporting Cancer (HTAN, AACR GENIE, MC2), Alzheimer's and Exceptional Longevity (AMP-AD, ELITE), Rare Diseases (NeuroFibromatosis, AMP-ALS), and Digital Health programmes.

Information Commissioner's Office

LONDON, UNITED KINGDOM

Head of AI & Data Science - Technology, Innovation & Enterprise Directorate, ICO

May '22 – Feb '24

- **Strategic Leadership:** Founded and grew the AI & Data Science Technology Policy team, transforming it into a national hub for regulatory policy development with direct influence on national AI governance policies. Pioneered strategies that deftly balanced technological advancement with data privacy and ethical AI use, setting new standards in regulatory frameworks.
- **Policy Innovation with Commercial Impact:** Co-authored ICO's Guidance on AI and Data Protection and the AI Risk Assessment Toolkit, establishing frameworks adopted by organisations across public and private sectors. Provided strategic guidance to high-value healthcare data initiatives, including Our Future Health, NHS England's FDP (Federated Data Platform), Great Ormond Street Hospital DRIVE and TREs (Trusted Research Environments), directly influencing multi-million pound programmes.
- **Cross-Sector Collaboration:** Engaged with government bodies to shape AI legislation, fostering collaborative approaches across regulatory agencies and international Data Protection Authorities. Built relationships across industry to champion Privacy by Design, providing practical guidance that balanced innovation with compliance.

Health Data Research UK, DARE UK, Visiting Scholar EMBL-EBI

LONDON, UNITED KINGDOM

Chief Technology Officer & Dir. of Engg - HDR UK & Technical Lead - DARE UK

Jan '19 – Jul '22

- **Portfolio & Financial Oversight:** Managed £69M+ portfolio of Health Data Research infrastructure projects, with direct accountability for budget allocation, milestone delivery, and financial reporting to funders. Oversaw multi-year programme budgets, ensuring efficient resource utilisation and demonstrating clear return on investment to government and charitable funders. Led commercial strategy for the HDR Innovation Gateway, developing sustainable operational models/
- **National Leadership:** Built and led teams of 200+ in-house and external specialists, establishing a customer-centric service delivery model. Established the national Trusted Research Environments (TREs/SDE) network, creating federated infrastructure connecting 100+ data custodians serving 45,000+ researchers. Achieved consistent high performance against SLAs and KPIs while maintaining

regulatory compliance.

- **Strategic Development & Partnerships:** Led multi-vendor digital transformation through the Health Data Research Innovation Gateway and DARE UK. Developed strategic partnerships with NHS Digital, Office of National Statistics, UKRI, and international bodies to ensure interoperability and secure data sharing. Championed Data Use Registers for transparency, establishing trust frameworks now foundational to the national health data strategy

EMBL - European Bioinformatics Institute

CAMBRIDGE, UNITED KINGDOM

Global Strategic Portfolio Manager - TSI, ELIXIR, EOSC-Life, EOSC-Hub & GA4GH Dec 17 – Jan 20

- **Major Programme Leadership:** Led the €26M EOSC-Life and €33M EOSC-Hub European and international GA4GH & HDR-UK infrastructure programmes, managing complex multi-stakeholder budgets across 50+ partner institutions. Delivered strategic technology projects including ELIXIR Cloud & AAI Analytics Platform, demonstrating capability to manage large-scale, multi-national funding. Co-authored EMBL-EBI's Hybrid Cloud Strategy report, establishing frameworks for cost-effective cloud resource utilisation.

- **Technical Coordination (June 2017 – November 2017):** Led strategic vision and technical roadmap for Human Data Communities in ELIXIR. Orchestrated €19M+ ELIXIR EXCELERATE initiatives, aligning technical platforms with scientific objectives. Contributed to successful grant proposals and pioneering applications of FAIR principles in pharmaceutical datasets.

University of Sheffield

SHEFFIELD, UNITED KINGDOM

CTO - Translational Technology Officer - CISTIB, Scientific Workflows Coordinator - VPH-Share, Insigneo Institute Jan 10 – Mar 17

- **Budget & Programme Management:** Architectural design and delivery of €18M VPH-DARE@IT hybrid cloud platform, managing budget across multiple funding streams and partner institutions. Development and operation of €13.4M VPH-Share cloud platform, demonstrating financial stewardship of complex distributed infrastructure. Coordinated 25+ research projects with combined budgets exceeding €30M, ensuring deliverables aligned with funder requirements.

- **Technical Leadership:** Managed GIMIAS biomedical image processing workbench and research-as-a-service capabilities for 70+ multi-modal clinical datasets. Fostered interdisciplinary collaboration across academia, industry, and healthcare partners

Please refer to my LinkedIn profile for a more complete list of work experiences along with recommendations.

Education

University of Milan-Bicocca

MILAN, ITALY

Executive MBA - Management of Research Infrastructures

2018 – 2020

Strategic, financial, and operational competencies for leadership of international research infrastructures. Focus on business models, sustainability planning, and cross-border governance

University of Sheffield

SHEFFIELD, UNITED KINGDOM

Doctor of Philosophy in Computational Systems Biology & Pure Mathematics

2005 – 2010

Hybrid, Hierarchical Models of Cardiac Cells and Tissues. ORS & Departmental scholarship, WUN Travel grant. SET Poster at House of Commons. Co-developed FLAME and CHASTE Frameworks.

Master of Science (Engineering) in Advanced Software Engineering

2004 – 2005

Distinction. Recipient of the Fretzwell-Downing Award for the Best Dissertation.

Pondicherry University

PONDICHERRY, INDIA

Bachelor of Technology in Information Technology

2000 – 2004

First class (86%) with Honors - Double Dissertation

Fellowships, Grants, Awards, Certifications, Board Memberships & Honours

Member of the UKRI Infrastructure Advisory Committee

2024 –

Scientific and Industry Board Member - Genomic Data Infrastructure (1+NG) project & MONARCH Initiative

2024 –

Sponsor - Digital Regulation Cooperation Forum, BSI/ISO ART/1 Committee Member

2022 – 2024

Fellow of the British Computer Society & Member of the Association of Computing Machinery

2021 –

FitSM - Advanced Service Planning and Design & Advanced Service Operation and Control

2019 –

Azure for Research Grant - \$40,000, AWS in Education - £16,000, WANDisco - \$60,000

2013 - 2016

Lifetime Fellow of Software Sustainability Institute & Member of the SocRSE

2015 –

Worldwide Universities Network Travel Grant & SET for Britain Poster at House of Commons

2006

Fretzwell-Downing Award for Best MSc Dissertation, ORS & Departmental PhD Scholarship

2005

Executive Leadership & Strategic Expertise

C-suite data, governance & technology executive with a track record of translating complex scientific, data, AI and technical capabilities into organisational strategy and measurable outcomes. Combines visionary thinking with operational rigour and equally comfortable presenting to boards and regulators as leading technical architecture decisions. Proven ability to build and scale high-performance teams across jurisdictions, manage multi-stakeholder relationships, and deliver transformational programmes in highly regulated environments.

Deep technical foundation spanning cloud infrastructure (AWS, Google Cloud, Azure), AI/ML systems (PyTorch, TensorFlow), and healthcare data platforms (TranSMART, REDCap, OMOP), ensuring credibility with engineering teams and the ability to make informed strategic technology decisions. Fluent in both Agile delivery and enterprise governance frameworks.

Selected Publications, Book Chapters, Conferences, Invited Talks & Research Reports

- M Akhtar, [...], S Varma, [...] *Croissant: A Metadata Format for ML-Ready Datasets*, NeurIPS, 2024
- S Ignatidou, [...], S Varma, *How do we ensure fairness in AI?*, Information Commissioner's Office, May 2023
- A Pearson, [...], S Varma, *AI & Data Protection Risk Toolkit*, Information Commissioner's Office, May 2022
- S Varma, T Hubbard, D Seymour, *Building Trusted Research Environments - Principles and Best Practices; Towards TRE ecosystems*, Dec 2021
- H L Rhem, [...], S Varma, E Birney, *GA4GH: International policies and standards for data sharing across genomic research and healthcare*, Cell Genomics, 2021
- J Lawson, [...], S Varma, M Courtot, *The Data Use Ontology to streamline responsible access to human biomedical datasets*, Cell Genomics, 2021
- Keynote Speaker at joint **European Commission & WHO** session on FAIR Data for future pandemic preparedness, 2021 & International Data Week 2022
- HDR UK, *COVID-19 HDR UK Research and Analysis reports to SAGE 2020-2021*
- N Karrar, [...], S Varma, *Analysis of Data Use Registers published by health data custodians in the UK*, (Preprint)
- A Wood, [...], CVD-COVID-UK Consortium, *Linked electronic health records for research on a nationwide cohort of more than 54 million people in England: data resource*, BMJ 2021
- G Saunders, [...], S Varma, S Scollen, N Blomberg, *Leveraging European infrastructures to access one million human genomes by 2022*, Nature Reviews Genetics
- M Fiume, [...], S Varma, S Scollen, *Federated discovery and sharing of genomic data using Beacons*, Nature BioTech, 2019
- B Gruening, [...], S Varma, D Blankenberg, R C Jimenez, BioContainers Community, Y Perez-Riverol *Recommendations for the packaging and containerizing of bioinformatics software*, F1000Research, 2018
- M Cabili, [...], S Varma, Pandya R, *Library Cards: Simplifying research access to genomics and health data*, Scientific Data, 2018
- P Morris, [...], S Varma, P Lawford, R Hose, J Gunn, *Fast Virtual Fractional Flow Reserve Based Upon Steady-State Computational Fluid Dynamics (CFD) Analysis: Results from the VIRTU-fast study*, JACC: Cardiovascular Interventions, 2017
- M Kasztelnika, E Coto, M Bubaka, M Malawski, P Nowakowski, J Arenas, A Saglimbeni, D Testi, A Frangi (Acknowledgement) *Support for Taverna workflows in the VPH-Share cloud platform*, Computer Methods and Programs in Biomedicine, 2017
- L Guo, [...], S Varma, A Venneri, A Frangi, Y Ventikos, *Subject-specific multiporoelastic model for exploring the risk factors associated with the early stages of Alzheimer's Disease*, Interface Focus, Royal Society Special Issue, 2017
- M De Marco, [...], S Varma, A Frangi, A Venneri, *ApoE ϵ 4 allele related alterations in hippocampal connectivity in early Alzheimer's disease support memory performance*, Curr. Alzheimer Research, 2017
- S Varma, VPH-DARE@IT Development Team, *The VPH-DARE@IT Platform for Translating Research to Clinical Decision Support of early and differential diagnosis of dementia*, VPH 2016
- D Warriner, A G Brown, S Varma, et al., *Closing the Loop: modelling of heart failure progression from health to end-stage using a meta-analysis of left ventricular pressure-volume loops*, PLOS-One, 2014.
- D Silva-Soto, S Varma, S Wood, R Hose *Workflows: Principles, Tools and Clinical Applications*, Book Chapter 8, Computational Biomedicine, Oxford University Press, 2014
- S Varma, [...], R Hose, *VPH-Share: Patient-Centred Multi-scale Cloud-Enabled Computational Workflows*, VPH-2014
- I M M van Leeuwen, [...], S Varma, [...], H M Byrne. *An integrative computational model for intestinal tissue renewal* Cell Prolif. 42:617-636, 2009.
- R H Clayton, [...], S Varma, [...], P Taggart, *Epicardial mapping of ventricular fibrillation in the human heart during ischaemia and reperfusion*. Computers in Cardiology 2008

- S Varma, M Holcombe *Extreme Programming: The Genesys Experience* eXtreme Programming 2005
- S Varma, M Holcombe, *Shared Code Repository: A Narrative* eXtreme Programming 2005

Please refer to my [ORCID:0000-0003-1687-2754](https://orcid.org/0000-0003-1687-2754) profile for a more complete list of scientific research outputs

Open Source Software, International Standards & Policies Contributions

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| • 2024-: Synapse.org, Synapse MCP Server | • 2024-: Synapse Knowledge Portals |
| • 2023-2023: How do we ensure fairness in AI? | • 2022-2023: Guidance on AI & Data Protection |
| • 2021-2022: GA4GH Computational Cohort Representation | • 2020-2022: HDR Innovation Gateway, Github |
| • 2020-2022: HDR UK Schemata | • 2020-2022: HDR UK Datasets, Papers/Preprints |
| • 2020-2022: HDR UK Phenotype Library | • 2020-2022: HDR UK Clinical Trials |
| • 2018-2020: GA4GH TRS, WES, TES & DRS Standards | • 2018-2020: EMBL-EBI ECP & AAP |
| • 2018-2020: ELIXIR TESK & WES-ELIXIR | • 2018-2020: EMBL-EBI RDSDS |
| • 2017-2020: GA4GH Beacon Network | • 2017-2020: Bioschemas Specification |
| • 2015-2017: VPH-DARE@IT Platform | • 2015-2017: GIMIAS Biomedical Workbench |
| • 2014-2015: VPH-HF Hypermodelling Framework | • 2011-2017: VPH-Share Cloud Platform |
| • 2006-2009: CHASTE | • 2004-2010: FLAME |